Implementing the Quality Function Deployment Method

Table of Contents

- Welcome to Your Course
- Read: Preview Your Course Project

Module Introduction: Relate System Performance to Design

- Read: Connecting Design Targets with Customer Needs
- Watch: QFD Terminology
- Read: Structure of the QFD
- Tool: QFD (House of Quality) Template
- Tool: Steps to Build a House of Quality
- Watch: Adding Performance Criteria to Your QFD
- Activity: List Your Performance Criteria
- Watch: Including Performance Criteria Weights and Optional Information
- Activity: Complete the Front Porch of Your QFD
- Watch: Rating the Performance of Your Current System, Options, and Competitors
- Activity: Determine Performance Scores
- Watch: Considering Back Porch Alternatives
- Discussion: Review and Improve Your Current System Concept
- Watch: Listing Your System's Engineering Characteristics
- Watch: Adding Engineering Characteristics to the QFD
- Activity: List Engineering Characteristics
- Watch: Relating Engineering Characteristics to System Performance
- Activity: Estimate Engineering Characteristic Impacts

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- Watch: Accounting for Interaction between Characteristics
- Activity: Notate Engineering Characteristic Tradeoffs
- Assignment: Course Project, Part One—Relate Performance Criteria and Engineering Characteristics
- Module Wrap-up: Relate System Performance to Design

Module Introduction: Define and Refine Design Targets

- Watch: Determining the Relative Importance of your System's Characteristics
- Activity: Calculate Imputed Importance
- Watch: Using Competitor Data to Inform Design Targets
- Activity: Establish Units, Constraints, and Competitor Values for Engineering Characteristics
- Watch: Assessing Technical Difficulty and Costs
- Activity: Re-evaluate after Technical Difficulty and Cost Considerations
- Watch: Establishing and Iterating Design Targets
- Activity: Assign Design Targets
- Watch: Using QFD to Resolve Counterintutive Results
- Discussion: Design Target Justifications
- Assignment: Course Project, Part Two—Quality Functional Deployment
- Module Wrap-up: Define and Refine Design Targets

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Welcome to Your Course

Video Transcript

You have a great understanding of your customers needs. And you have well-defined performance metrics to evaluate how any solution meets those needs. But now, you have to decide, well, if your customer wants it to be fast, how fast is fast? Or if they want to be durable, what are all the different kinds of engineering parameters that actually influence whether or not it's durable. What should be the design targets for those?

Quality functional deployment or QFD is one of the most effective tools for relating performance metrics that your customer cares about to the technical criteria and engineering parameters your team needs to determine design targets for in order to build their solution. And it does so in a way that you can compare your concepts to your competitors, understand the trade offs between engineering parameters and their influence on performance criteria, and ultimately, be able to defend why your design targets will lead your team to a winning solution.

In this course, we're going to help you go through a step by step process in building your own QFD all the way through, taking your performance criteria and evaluating against competitors to determining lots of different kinds of engineering characteristics and their influence on the performance criteria, the interrelationship between those engineering characteristics and ultimately, be able to use all that information as well as its technical difficulty, it's cost involved with these in order to establish really strong design targets and get an estimate as to what your final systems performance will be.

Course Description

One of the most challenging aspects of system engineering is figuring out how best to manage system performance trade-offs. Time, budget, and other limiting resources force you to prioritize aspects of your design so your system provides the maximum benefit. You can manage these trade-offs efficiently and effectively by implementing a process known as Quality Function Deployment method, or QFD.

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QFD is also sometimes referred to as "house of quality," and the document that is created during QFD does in fact look like a house. The house of quality provides a crucial visual link between your emerging system design, the needs of your customers, and competing designs in the marketplace. With QFD you'll build intuition about your system, make decisions about resource allocation, communicate development priorities, and establish design targets, all in a way that ensures you are creating the best-performing solution possible. Design teams of all types use QFD, and it's one of the most established and recognized systems design tools today.

In this course, you'll be led through a detailed, step-by-step process to build a house of quality for your own project. This process begins with identifying what your system must do for customers (performance measures) and which of these performance measures matter most to the customer. You'll relate each of your system's engineering characteristics to those performance measures. After this, you'll complete the house of quality by considering each engineering characteristic in terms of importance, difficulty, associated cost, the current competitive benchmarks, and minimum or maximum requirement thresholds. This process will lead you to performance targets that consider both what is desired and what is feasible. The collection of design targets will give you a good working estimate of the best performance you can expect your system to achieve.

What you'll do

- Define key engineering characteristics
- Determine impact of each characteristic
- Benchmark against competitors
- Determine initial targets for each engineering characteristic

Faculty Author

David R. Schneider earned his Masters and PhD from Cornell University in Mechanical Engineering with a concentration in Controls & Dynamics in 2007. David has taught at both Cornell and Columbia University. David Schneider's research has traditionally focused on the realm of NP-Hard Computer Science Problems and Controls for Robotic Systems in centralized and

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David R. Schneider **Director of MEng Studies in Systems Engineering** Cornell Engineering Cornell University	decentralized as well as autonomous and semi-autonomous systems. His most prominent research is his creation of the G*TA (G-Star-T-A) task allocation algorithm and his work as Program Manager of the Cornell RoboFlag program, with notable applications including AFRL UAV controls and NASA/NOAA unmanned boat designs. With a strong focus on education, David's endeavors have included the creation of the Cornell Cup, Innovative Embedded Design National Competition; leading Cornell University Sustainable Design (CUSD); and the broader impacts video game creation for the NSF Expeditions in Computing Grant on Computational Sustainability. David has led the efforts to make Cornell the first university to officially partner with Make: and is a leader in the Higher Education Maker Alliance working with the White House Office of Science and Technology Policy.

Back to Table of Contents

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Read: Preview Your Course Project

Throughout this course, you will practice applying the teachings of each module in a multi-part course project. The final project part, in which you will synthesize all course findings, is worth a significant percentage of the whole. You can review all relevant information on the **Grades** page of this course.

☆ **Key Points**

You will apply the content from this course in a multi-part project.

You must submit all prior assignments to access the final project part.

Since each module builds on the previous ones, it is essential that you work through the course in the order it appears. You must submit all prior assignments to access the final project part.

Course Project Parts

Preview the course project below. As you work through the course, reflect on how this relates to the course content and to your experience.

Note: Though your work will only be seen by eCornell, you should take care to obscure any information you feel might be of a sensitive or confidential nature.

– **Course Project, Part One — Relate Performance Criteria and Engineering Characteristics**

In this module, you'll examine how filling out the House of Quality brings key project data into one unifying framework and allows you to manage trade-offs among the performance requirements and characteristics of your system. The main task in this part of the course project is using the House of Quality (QFD) Template to begin to build out your own QFD. Your project document will guide you through the requirements, but in essence you need to build out:

- the front porch (performance criteria with their weight of importance)
- the back porch (normalized scores, both weighted and unweighted, for your system, your competitors, and any other systems options you wish to

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consider)
- the second floor (list of engineering characteristics as well as the direction of change arrows)
- the main floor (matrix that identifies the impact of engineering characteristics on performance criteria)
- the roof (matrix that identifies the impact of engineering characteristics on other engineering characteristics)

Course Project, Part Two — Quality Functional Deployment

In this part of the project, you will build out the basement of your House of Quality. In the basement you will collect additional information about your engineering characteristics, including:

- the set measurement units
- the engineering values of your competitors
- any external requirement thresholds
- the imputed importance
- any technical difficulty
- any estimated cost

With all this information established, you then will provide educated estimates toward the **design targets** for your system.

In addition to filling out the basement, you will create a new column on the back porch and fill it out with a set of estimated scores based on a system that is designed with the targets you have just added in the QFD basement.

Back to Table of Contents

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Module Introduction: Relate System Performance to Design

The Quality Functional Design (QFD) begins with the customer needs, and these needs are expressed using a list of attributes that will be used to measure system performance. Basically, for each of the important criteria (reliability, accuracy, portability, safety, etc.) you're using a performance measure to express how well your system scores. In addition to a scaled score for performance, each of the criteria are weighted for importance relative to the other criteria. This lets you focus on what matters most to customers while still accounting for all important criteria.

With broad performance measures established, you then need to turn your attention to the actual nuts and bolts of your system. For example, how might your system's mass affect criteria like portability and safety? By focusing on specifics, you can connect the detailed characteristics of your system to a measurement of how well it will meet customer's needs. This is a key outcome from using the QFD method, and making this connection will be the focus of your work in this module.

In this module, you'll begin to build your house of quality. You'll start by establishing performance criteria and determining their relative importance with respect to customer needs. You'll use these criteria to evaluate your system's performance against competitors' performance. Next you'll identify all engineering characteristics of your system and assess their impact on the performance criteria. Finally, you'll examine interrelationships between engineering characteristics so you can better understand how tradeoffs might impact your design.

Back to Table of Contents

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Read: Connecting Design Targets with Customer Needs

Every design involves trade-offs.

Here are some examples of trade-offs:

- Do we increase the maximum speed, even if this increase means using more fuel and reducing the run time of our system?
- Do we make our system more expensive, or is it better to re-allocate funds from one subsystem to another with the possible result that one or more subsystems will not perform as well?
- Do we require a low error rate, or increase the allowable error rate and make our system handle errors more easily?

☆ **Key Points**

The QFD method brings into focus the interplay between system performance, customer needs, and specific design targets.

A good approach to setting design targets takes into account both the impact on performance measures and the potential impact on other design targets.

Throughout the design process you need to address these kinds of trade-offs in a way that results in the best system—and in a way that you can justify to others. The design decisions you make along the way will result in specific design targets. These design targets are desired values that you set for each of your system's engineering characteristics, such as how fast your system can go or how much it weighs. But how do you know that the design targets you've determined are appropriate?

In order to show that your design targets will result in a system that meets customer's needs, it is important that you:

- define performance criteria for your system
- relate each of the performance criteria to customer needs
- establish the relationship between the performance criteria and each engineering characteristic
- understand how adjustments to design targets for each engineering characteristic will influence performance measures; and
- understand how adjusting one design target may also influence the ease of meeting other design targets

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This method, the Quality Functional Deployment (QFD) method, is implemented by creating the “House of Quality”, which is a document that ties together system performance, customer needs, competitive analysis, and design targets. The unique value of the visualization is that all these important factors in the design process are together in one place and linked so the potential effects of any given change can be seen more easily by the whole team.

Back to Table of Contents

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Watch: QFD Terminology

When designing your system, you need design targets. But what exactly are design targets? You may know they have to do with engineering characteristics, but what are engineering characteristics? And how do they relate with QFDs?

In this video, Dr. Schneider defines these key terms used within the industry and shares how these terms relate to the design of your system.

Video Transcript

Whenever you start to learn about a new systems engineering tool, it's always important to make sure you're familiar with the terminology to make sure that you really come across professionally. Now, the QFD is really all about design targets. So what are design targets? They are the intended value or desired value for any engineering characteristic.

Okay, so what's an engineering characteristic? Engineering characteristics is a term that we use to generally refer to any property of your system that you have control over. Maybe it's speed, it's weight, it's size, it's mean time between failure, it's response time, material cost, manufacturing costs, basically any kind of knob in your design that you can tweak is referred to as an engineering characteristic. The issue often is, of course, that adjusting one of these knobs might affect the others so maybe adjusting the size might also affect the weight property of it as well. Now, both of those: the size and weight are also going to have an influence on the performance of your overall system. And will show how the QFD is a way to represent these kinds of relationships and trade-offs overall.

Just for completeness, the QFD is also called the House of Quality quite often due to the shape of its matrix. If you use your imagination it kind looks like a house and as we'll see different parts of the matrix are given common names that refer to parts of a house as well.

Back to Table of Contents

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Read: Structure of the QFD

A carefully constructed QFD will help you navigate the interplay between your customer needs and your engineering characteristics, and it will help you arrive at optimal design targets for your system. The matrix ties together so many essential pieces of information in a single, organized structure, that this diagram is often referred to as one of the master documents of systems engineering.

☆ **Key Points**

A QFD brings together key information in a single, organized structure.

The QFD is built in a step-wise fashion, beginning with performance metrics and priorities that reflect customer needs.

When you first look at a House of Quality diagram, though, it may seem overwhelming. Don't panic!

It's helpful to begin by looking at the structure of the diagram and seeing how it resolves into just six blocks of information. For now, don't worry if you're not exactly sure what is indicated or required in each block. This course will make QFD development clear and attainable by guiding you through each step in detail.

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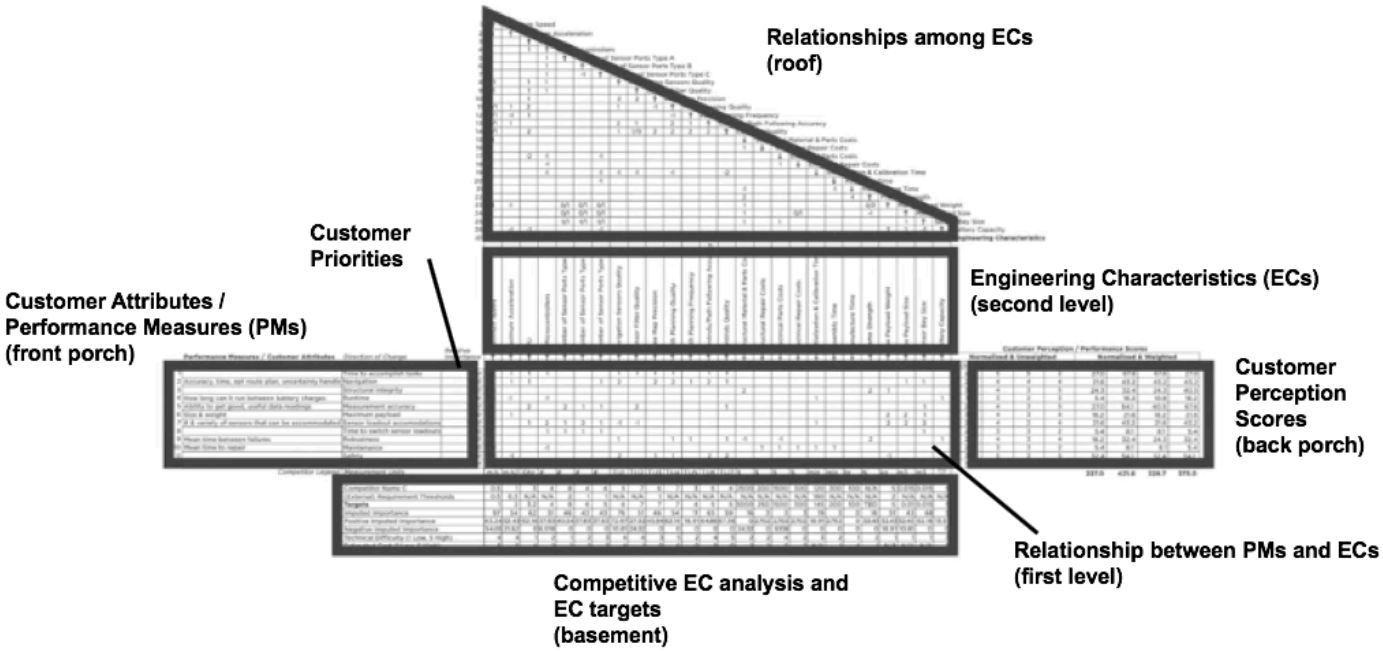

These are the sections of the QFD, roughly in the order in which you'll typically populate them:

1. Customer Attributes / Performance Measures (PMs) / Customer Priorities
2. Customer Perception Scores (for your system and for competitors)
3. Engineering Characteristics (ECs)
4. Relationship between PMs and ECs
5. Relationship Among ECs
6. Competitive EC analysis and EC targets

Notice that focus on the customer is heaviest at the beginning of the QFD process. It's important to lock in the performance measures that matter to the customer early to make sure the work that follows is helping you design not only the best system but (just as importantly) the right system.

Back to Table of Contents

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Tool: QFD (House of Quality) Template

Download the Tool

QFD (House of Quality) Template

This is a template you can use as a starting point when you need to build a QFD, or House of Quality. The template has a preset format that you can modify, adding and deleting rows and columns as necessary. It also includes a completed sample that you can use as reference.

Download this template and keep it handy for any project in which you'll be building out a QFD.

Note: This tool is a Microsoft Excel workbook. One sheet includes the template you can use as a resource to create your own QFD. The second sheet shows a sample QFD developed for a robotic vehicle. As you watch the videos, you can follow along using this provided sample. The third sheet is a reference that defines basic robotics engineering characteristics and gives other information to help you better understand the robotic vehicle example.

Back to Table of Contents

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Tool: Steps to Build a House of Quality

Download this checklist and use it when you need to create a House of Quality (QFD). You will use this checklist to build out your own QFD as you go through the course. At the completion of these steps, you will have a well-defined way to compare design targets and show how the adjusting of design targets influence other design targets.

Download the Tool

Steps to Build a House of Quality (QFD)

Note: The steps have been taken from the **Determining Design Targets Guide**. This course will take you through each of these steps. However, before you begin, you are encouraged to read through all the steps to get an overview and see how they all fit together. With the big picture in mind, this will help you perform each step better when you create your own QFD.

During the process, you may find that you want to perform some of the steps in a different order. This is acceptable as long as

- you have a clear vision of the end goal, and
- when the QFD is completed you have combined all factors so that the results can be correctly interpreted

Back to Table of Contents

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Watch: Adding Performance Criteria to Your QFD

In this video, Dr. Schneider begins to fill out a sample QFD, starting with the performance criteria on the "front porch" of the diagram. He emphasizes that this is a working list and does not necessarily need to be comprehensive to start.

To follow along, open the **House of Quality (QFD) Template** and take a look at the *QFD Sample for Robotic Vehicle* sheet. Here you'll see performance criteria listed in the leftmost section of the QFD.

Video Transcript

Before you begin your QFD work, you should have your performance criteria and metrics already established. As these can vary for each project quite significantly, it's helpful for us to have a sample product for us to talk about. And let's pick one that's got some real meat behind it, to give us something to really dig into.

So let's imagine you're creating a semi-autonomous rover, some kind of little robot that's meant to take various sensor readings as it travels along. Let's also imagine that the sensors used in one set of tasks may be different than another. And hence we may need to be able to swap out of some of our sensors. So we might come up with a list of criteria like those in the example. Now there may be more criteria than what's listed here in the example, especially depending upon the kind of vehicle, the environment, kinds of tasks it's expected to handle et cetera. But this is a good list for us to work with for our purposes overall.

Now to start your QFD, you simply take your list of criteria and simply place one criteria in each row. This is often referred to as filling out the front porch of your House of Quality. Now we'll be adding a lot of information to our House of Quality, so we highly recommend that you make use of the template. Now, when looking at the template, please don't panic. I know it may look real large and unwieldly, but it ties together so much great information, that in the end, you may even find it to be a little bit elegant. In fact, it is so useful being able to pull together all this information, it's often referred to as one of the master documents of systems engineering.

Back to Table of Contents

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Activity: List Your Performance Criteria

In this activity, you populate the **House of Quality (QFD) Template** with performance criteria identified for a project that you are currently working on or one that you completed in the past. If you don't have a project you can readily use, consider using the attached **QFD Scenario** for a laser tag toy.

Your work in this activity and the subsequent activities in this course will culminate in the completion of the Course Project, which you'll submit to your instructor as part of the course completion requirements.

Task: List Your Performance CriteriaTask: List Your Performance Criteria
This is Complete When...This is Complete When...
Worked ExampleWorked Example
Back to Table of Contents

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Watch: Including Performance Criteria Weights and Optional Information

When you look down your list of performance criteria, you probably notice that some seem more critically important than others. You need to assign weights to each of the criteria so that as you make calculations later in the QFD process you are accounting for the relative importance of each of your criteria.

Since your QFD will be used as a communication tool throughout the project and across your team, it will be useful to include additional descriptions for some of the performance criteria. This additional information can be provided as an optional extension of the QFD.

In this video, Dr. Schneider describes where to include relative importance values and additional information in your QFD.

Video Transcript

As a QFD is very popular and powerful tool, there have been some common variants. Sometimes, you'll see next to the performance criteria, brief descriptions of each performance criteria listed in the leftmost column. As performance criteria focus on the evaluation of meeting customer's needs, there may be additional needs or criteria that you as a designer may also want to consider. Some common ones may include manufacturing assembly time, commonality and parts with other company products, profit margin. Things like this that again, you as a designer care about but maybe not again really touch upon the customer's needs as much.

Now, these additional criteria are often listed below the customer's performance criteria and are often italicized in order to help distinguish them. For all criteria though, it is very important to also record the weight of the criteria. This is often labeled as the relative importance. And this is almost always listed as a percentage in the column that is just to the right of the criteria.

Back to Table of Contents

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Activity: Complete the Front Porch of Your QFD

In this activity, you will assign weights to your criteria to reflect their relative importance. You will also identify which criteria could benefit from extra information and add this information to the QFD. Finally, you'll review your performance criteria and add any criteria you might want to use for internal tracking only.

Review the worked example to get a better idea of how to work through the details of this activity.

Task: Add Additional Details for Performance CriteriaTask: Add Additional Details for Performance Criteria
This is Complete When...This is Complete When...
Worked ExampleWorked Example
Back to Table of Contents

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Watch: Rating the Performance of Your Current System, Options, and Competitors

You, your team, and your customers all want to know how well your system will perform. The answer to that question will be based on the performance criteria you've defined and weighted. Once you have established this basis, you're ready to rate your system performance. Of course, this rating can't be absolute. It needs to be comparable to something else. Competitors' systems are an obvious choice for comparison. But you may also have more than one candidate designs for your system, and you should consider rating against these alternative options as well.

In this video, Dr. Schneider describes how to determine performance scores for your design, your competitors' systems, and any alternative candidate designs. He also emphasizes the importance of recognizing who your competitors are at this stage of your design process.

Video Transcript

With performance criteria established, a common next step is to use that performance criteria to evaluate your current solution, possibly multiple options you're considering, as well as the quality of any competitors to your current solution. Now, one way to do this, and a way that is often preferred professionally by many people including myself, is to essentially incorporate decision matrix into the back porch of your House of Quality.

Now one of the common mistakes or shortcomings that people often have in doing this is they don't include enough competitors. Unfortunately quite often people might have the attitude that there are no competitors. What we are doing is completely novel in what we do, but trust me there are always competitors. People have likely had the need you're addressing before your project came along, or at least part of that need, and had been looking for ways to meet that need.

So let's take a very simple example to demonstrate this. So let's imagine that you came up with some kind of pirate ship toy with shooting cannons. And let's also pretend that for whatever reason, there has never been a pirate ship toy before. So you really do feel that it actually is novel. However, there have still been many other solutions that have probably met the same customer needs that your pirate ship toy has. Here's a simple

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example, maybe pirate video games if you will. They are able to create some kind of a game play experience around the pirate mentality, right. So maybe that's a key need that you're going for. Maybe it's more the storytelling element that you can tell with a pirate ship, so maybe pirate movies are a good competitor to consider. Maybe it's actually the role play of being a pirate that they like, and so maybe costumes can be a competitor. Maybe even just toy boats are a competitor or even just other toys that shoot. These are all different solutions out there that meet at least part of the need.

Now, granted they don't meet all the needs you might be trying to target, and hence for some of your criteria, those competitors might get simply a score of zero. But your customer is looking to meet their needs with other solutions. And if your customers are looking at it, you need to look at it too. It's also okay if your system is currently performing poorly, you want to be honest. This is actually really important to identify these weaknesses early on, so you will be able to know how to improve upon them later on after you've done your QFD analysis.

Back to Table of Contents

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Activity: Determine Performance Scores

In this activity, you will begin by rating your system's performance using the performance criteria you've identified in your **House of Quality (QFD) Template**. After determining how your system scores, you'll need to weight the scores. You'll then rate your competitors' solutions and any other options you're considering for your system. All this work will be done on what's referred to as the back porch of your QFD.

Review the worked example to get a better idea of how to work through the details of this activity.

Task: Find the Current Weighted Scores Task: Find the Current Weighted Scores
This is Complete When...This is Complete When...
Worked ExampleWorked Example
Back to Table of Contents

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Watch: Considering Back Porch Alternatives

In this course, the performance scoring displayed on the back porch of your QFD is based on a decision matrix method. This is a numbers-based comparison of your system options and your competitors. Some other methods in common use represent the options diagrammatically. These are considered easier to read, but they come with potential drawbacks.

In this video, Dr. Schneider describes two alternatives for back porch visualizations and exposes some of the possible shortcomings of each alternative.

Video Transcript

>> There are many kinds of visual representations that can be used for the back porch of your house of quality. Now decision matrices are my favorite way, and again considered to be a fairly professional one overall, but there are other ones. Traditionally, people use sliders in order to represent the normalized score for the various solutions, but this can get a bit crowded when you have multiple options or competitors. Another popular alternative is to use radial charts. These are very popular in parts of Asia in particular. However, these can sometimes be misleading. Look at these two examples of radial charts here. The one on the left looks like it's one sided. The other one looks a bit balanced overall, but they're actually presenting the same information. We just changed the order of the axes.

Back to Table of Contents

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Discussion: Review and Improve Your Current System Concept

It's important to be aware of weaknesses in your system. Where possible, you should try to address these weaknesses early in your design process while you still have the time and resources to make significant improvements.

In this discussion you and your classmates will share information about your systems and help one another identify opportunities for improvement. These interactions will give you a chance to see your QFD from a different perspective.

Instructions:

You are required to participate meaningfully in the course discussions.

Discussion topic:

Consider your performance criteria and the corresponding weights. Create a post in which you:

- briefly describe your system concept
- list your main competitors
- share screenshots of the front and back porch of your QFD
- indicate your competitors' strengths and weaknesses
- indicate how your system's strengths and weaknesses compare to your competitors

Once you have created your post, choose at least one post made by another student and create a reply in which you:

- ask clarifying questions about the system concept
- suggest any additional competitors you think they might want to consider
- share any lessons learned from reading the original post

To participate in this discussion:

Click **Reply** to post a comment or reply to another comment. Please consider that this is a professional forum; courtesy and professional language and tone are expected. Before posting, please review eCornell's policy regarding plagiarism (the presentation of someone else's work as your own without source credit).

Please note:

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While discussion board postings will be accepted through the end of the course, we strongly encourage everyone to move through this course as a group. As such, postings that are made to this board after 5pm ET on the due date will be read and graded but may not receive a response on the board from your facilitator and/or your peers. Please let your facilitator know if you have any questions.

Back to Table of Contents

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Watch: Listing Your System's Engineering Characteristics

While it's useful to understand the competitive landscape in which your system needs to perform, setting design targets that optimize your own system's performance is the main focus of the QFD. You need to look in detail at components and functions of your system and consider how each contributes to meeting the performance criteria. The engineering characteristics of your system capture the functional and structural aspects of your concepts over which you have control.

In this video, Dr. Schneider explains the difference between engineering characteristics and performance criteria. By working through an example, he gives insight into the methods used to generate a complete list of engineering characteristics.

Video Transcript

Now we're ready to create a list of engineering characteristics. Now remember, performance criteria focus on measuring any solution. Engineering characteristics are for your solution concept and can refer to both functional and structural aspects. Now quite often if you've already developed things like use cases, use case behavioral diagrams or even FFBDs reviewing those can be very informative for brainstorming engineering characteristics. But a lot can be determined by just considering your systems operation.

As an example of doing this, let's consider that semi-autonomous rover again and go over a brief description of its operation. Now the operation begins with the vehicle getting a command to go to a new destination and operate its payload and navigational sensors along the way. Now to perform this task, the vehicle has a suite of on-board navigation sensors that help it to estimate the world around it, and then populate a local area estimate or map, or you can even think of this as a grid, that says how safe it is to travel in each one of the grid cells. This map is then used by a path-planning algorithm to determine a route for the vehicle to travel from its current location to the desired destination. This vehicle then runs a low-level control algorithm to create commands for the vehicle's motors to drive the vehicle along that calculated path. And as the vehicle moves, new sensor information is brought in to update the map and the process basically repeats itself.

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Now, from this description we begin to derive a list of engineering characteristics. We can start to think about, what are the functions that must be achieved and what aspects of our system might affect how well we're able to achieve them. Let's take a very simple example recognizing that the rover needs to travel. And so some obvious engineering characteristics that might jump to mind are things such as, you know, what is the maximum speed or maximum acceleration that our vehicle has. Similarly, as the rover needs to be able to make a map, we might have engineering characteristics that focus on the quality of our navigation sensors or our ability to filter that sensor data. We might even have engineering characteristics that say, you know, how good of a map are we able to create from that data.

Now once we have that map created, we also have engineering characteristics that focus on the quality of our path-learning algorithm that we have and the ability of a robot to follow that path which might be like our controls, so that way our robot is actually able to transverse that world and transverse it hopefully very well. And then we can begin to really delve even deeper thinking about what aspects of our system might be related or even directly affect the engineering characteristics that we've already come up with. As an example, as we have to transfer this world with our path the frequency that we calculate our path could affect how quickly our robot's able to react. Similarly, we can think about how precise of a map are we going to have to create and that could impact the quality of our sensors and filters needed. Hence things like frequency and precision can also be very good things to include as engineering characteristics.

There are of course other things that our system needs to do such as carry and support various sensors of different types. And there are realistic design operational constraints to consider as well such as cost; maybe repair cost or repair time, maybe the assembly and manufacturing times, or initialization and calibration times. And there can be lots of other ones. In fact it's not uncommon to consider on the order of somewhere between 15 to even 50 engineering characteristics.

Now notice we're also using words like quality in describing our engineering characteristics, and we'll talk about how to characterize quality even for things like algorithms later on. But quality is often used as a general term and the more you're increasing the engineering characteristics quality, it's basically assumed the better results that you're able to provide. Even though it may take more or fancier resources in order to achieve or operate.

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Watch: Adding Engineering Characteristics to the QFD

Once you have identified what your engineering characteristics are, it is time to add them to your QFD and begin to think about how each engineering characteristic impacts the performance criteria.

In this video, Dr. Schneider describes how to populate engineering characteristics into the template. He also discusses how to set direction of change arrows for each engineering characteristic.

Video Transcript

To incorporate engineering characteristics into your QFD, the engineering characteristics are listed as a column shown in the template. This is often referred to as the second floor of a house of quality and note that is common to have the text actually rotated. Now, also notice that there is one row that is left open in the template just below the engineering characteristics. This is designated as the direction of change row, and it's basically for each engineering characteristic is given its own up or down arrow in this row.

If the arrow points up this means when we talk about adjusting this engineering characteristic we ask the question, "What impact will increasing this engineering characteristic have on our performance?" For example, since the arrow under our maximum speed is pointing up, we'll be asking the question in the QFD, "What impact will increasing the maximum speed have on the system's performance criteria?"

Now, there's actually no wrong way to pick the direction of change. However, as a rule of thumb the arrow is usually chosen to point in the direction believed to have an overall positive effect on your system performance. It doesn't have to though, but this is just a common practice.

Back to Table of Contents

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Activity: List Engineering Characteristics

In this activity, you will list your engineering characteristics in your **House of Quality (QFD)**. Then you will identify the direction of change for those characteristics.

Task: List Engineering CharacteristicsTask: List Engineering Characteristics
This is Complete When...This is Complete When...
Worked ExampleWorked Example
Back to Table of Contents

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Watch: Relating Engineering Characteristics to System Performance

Now that you have established your engineering characteristics, it's time to look at how these characteristics will impact the performance criteria.

In this video, Dr. Schneider describes how to use the matrix in your QFD to capture predictions of how a change in each engineering characteristic will affect performance.

Video Transcript

With both your performance criteria and your engineering characteristics included in your QFD, you've essentially created a matrix. This matrix is referred to as the main body or main floor of your house of quality. And what we're going to be doing now is filling in the cells by essentially asking the question "How does changing the engineering characteristics impact the performance criteria?".

Traditionally, these cells are filled in with integer values ranging from minus two to two. Positive numbers indicate that adjusting the engineering characteristic, according to its direction of change arrow, will have a positive effect on that performance criteria. For example, the direction of change for the maximum speed engineering characteristic is given an up arrow. Therefore, we now ask "We'll increase the maximum speed, what effect will that have on the performance criteria?". For example, taking it further, increasing the maximum speed will have a very positive effect on performance criteria of time to accomplish tasks. Therefore, that's how it's given a two. If it had a very negative effect on that performance criteria, it would've been given a minus two.

Other times you'll see that the cells are filled in with symbols. Common ones can include Xs or minuses for negative values, or checkmarks and plus signs or positives. In some cases you may even see people putting high medium low values or HML, numerically equivalent to integer values between minus three and three. However, make sure that you never go to a wider scale that minus three to three. Some people may even shun minus three to three only accepting minus two to two. Why is this? Well, when the QFD is commonly first used in design, the design concept is not fleshed out well enough nor the engineering characteristics influence on the performance criteria understood well enough to justify that much of distinction. So, do you really want to give something a five and that a five really means it's five times more

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influential than something is given a one? That's not something we can usually defend well at an early stage.

Instead, what we're really trying to do here is establish general trends. This way sometimes people even use the symbols to get away from the numerics. We're trying to actually just say whether something strongly, somewhat or not significantly, i.e., a zero, affects the performance criteria. Consequently, the QFD main body is often sparse with most cells being given a zero. Yes, there may be some small influence but the majority of the time we don't have enough information to accurately quantify this small influence, so we leave those cells blank.

Back to Table of Contents

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Activity: Estimate Engineering Characteristic Impacts

In this activity, you will populate the matrix on the main floor of your QFD with estimates of how adjusting each engineering characteristic will affect each performance criteria.

Task: Fill in the MatrixTask: Fill in the Matrix
This is Complete When...This is Complete When...
Worked ExampleWorked Example
Back to Table of Contents

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Watch: Accounting for Interaction between Characteristics

One of the greatest features of the QFD is its ability to compare multiple things at once. Since one of the essential considerations in establishing design targets is the trade-off required among engineering characteristics, you will want to utilize the roof portion of your QFD as a matrix in which you express the relationships between engineering characteristics.

In this video, Dr. Schneider describes how to populate the roof portion of your QFD and explains what this part of the diagram means for you and your team as you examine your concept's design.

Video Transcript

Now, we're going to compare engineering characteristics' effect on other engineering characteristics. We're going to represent this in the QFD roof, where the engineering characteristics are not only going to be our columns now, but also will have those same engineering characteristics listed as rows. This is going to be somewhat very similar to how we compare engineering characteristics influence to performance metrics. However, now we're going to recognize how about making trade-offs in design concept itself, while still using the same minus two to plus two scale.

As an example, basically, one comparison engineering characteristics might be increasing the maximum speed engineering characteristic can be shown as having a negative effect on battery capacity, because increasing the maximum speed would cause a larger drain on the batteries. Similarly, improving the CPU engineering characteristic would have a positive effect on many of the quality engineering characteristics, because a better CPU would make it easier to operate many of these complex functions.

Now, here these could be recorded in the reverse way as well, such as increasing the battery capacity could have a positive effect on your maximum speed. But you'll notice that as a result of many of these relationships stored in the QFD roof, are basically symmetric in this way. We often represent the QFD roof as being basically a symmetric triangular matrix. And now, notice I use the word often. Now, there are a few engineering characteristic relationships that are not symmetric. As an example, increasing the maximum speed would make the path planner and controls more

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difficult. Now, this would be a negative relationship, but improving the path planner and controls would allow the maximum speed to increase, which would be a positive relationship.

So, in these special cases the asymmetric relationship is written as the columns engineering characteristics effect, on the row engineering characteristic then a slash. Then, the rows engineering characteristic effect on the columns engineering characteristic. Now, should you have an asymmetric relationship where one direction is neutral, the neutral relationship is often represented as a zero. Or if you haven't been using minuses to represent negative marks in your QFD so far, you'll also see sometimes the neutral relationship represented by a dash. Now, this is shown in the example, at least the neutral relationship is shown in the example, where sensor filter quality will have a positive effect on controls quality, but controls quality doesn't really have a significant effect on sensor filter quality.

Now, with all these parts of the QFD filled out, now is a great time, actually an essential time to review this with your team. Do you agree that engineering characteristics impacts on the performance are as they are represented here? Do you recognize and understand all the trade-offs between those engineering characteristics? This should not be just a tacit okay with, but rather should really be an in-depth discussion. You should be willing to help the team to recognize the impact of their design decisions. Both towards the overall performance of the system as well as how their design decisions can aid, or hamper other teams part of the design.

Back to Table of Contents

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Activity: Notate Engineering Characteristic Tradeoffs

In this activity, you will populate the matrix with your estimates for how adjusting each engineering characteristic will affect each performance criterion.

Task: Adjust Engineering CharacteristicsTask: Adjust Engineering Characteristics
This is Complete When...This is Complete When...
Worked ExampleWorked Example
Back to Table of Contents

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Assignment: Course Project, Part One—Relate Performance Criteria and Engineering Characteristics

Throughout this module you have seen how filling out the House of Quality brings key project data into one unifying framework and allows you to manage trade-offs among the performance requirements and characteristics of your system. If you have been completing each step of the process along the way, completing this part of the course project should be very simple and straightforward.

The main task in this part of the course project is using the **House of Quality (QFD) Template** to begin to build out your own QFD. Your project document will guide you through the requirements, but in essence you need to build out:

- the front porch (performance criteria with their weight of importance)
- the back porch (normalized scores, both weighted and unweighted, for your system, your competitors, and any other systems options you wish to consider)
- the second floor (list of engineering characteristics as well as the direction of change arrows)
- the main floor (matrix that identifies the impact of engineering characteristics on performance criteria)
- the roof (matrix that identifies the impact of engineering characteristics on other engineering characteristics)

Completion of all parts of this project is a course requirement.

Instructions:

1. Download the **course project document.**
2. Complete Part One.
3. Save your work. This includes both the partially-completed QFD workbook and the course project document. Save your work as one of these file types: .doc, .docx, .txt, .pdf, .xls, .xsls. No other file types will be accepted for submission.
4. You will not submit your course project for grading and credit now. At the end of the course, you will submit your completed project for instructor review and grading.

Do not hesitate to contact your instructor if you have any questions about the project. You will add to this document as the course proceeds and will submit it to the course

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instructor at the end of the course.

Before you begin:

Before starting your work, please review the **rubric** (a list of evaluative criteria) for this assignment. Also review eCornell's policy regarding **plagiarism** (the presentation of someone else's work as your own without source credit).

Back to Table of Contents

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Module Wrap-up: Relate System Performance to Design

In this module, you defined key performance criteria and determined relative importance for each of these criteria. You established performance scores that you can use to compare your system against competitors and other possible design options. You established the engineering characteristics of your system and considered their impacts on these criteria and on one another. At this point, most of your House of Quality seems to be filled out.

The work you've done so far establishes your system's value in relation to your competitors, and it allows you to consider trade-offs in your system's design characteristics. You're now ready to complete the House of Quality by analyzing all the information you've gathered so far to establish design targets.

Back to Table of Contents

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Module Introduction: Define and Refine Design Targets

Now that you've begun to get a solid grasp on your performance criteria and engineering characteristics, you will want to tackle the basement portion of your QFD. This area of the diagram deals with calculated imputed importance, technical difficulty assessments, and informed design targets.

In this module, you will build out the basement portion of your QFD by determining initial targets for design parameters. You also will develop guidelines for resources allocation that supports those targets.

Back to Table of Contents

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Watch: Determining the Relative Importance of your System's Characteristics

Not all engineering characteristics are equally influential on your system's performance, and your QFD is designed to help you recognize this. There is a way to calculate how much influence each engineering characteristic has on your potential solution's overall performance. The influence of each characteristic is expressed as a percentage, and these percent values are described using the term imputed importance. You can calculate imputed importance simply using values you've already input in your QFD.

In this video, Dr. Schneider describes how to calculate the imputed importance and enter it into the basement. You may find it helpful to look at the QFD Sample tab in the provided template to see how these values are calculated.

Video Transcript

With the body of your QFD established, it's worthwhile to try to summarize the importance of each engineering characteristic in terms of its impact on performance criteria. Now, this is the engineering characteristics imputed importance. Now, to begin to calculate this, you want to translate your QFD's body into absolute integer values. So, your Xs and pluses become minus two to two. Then, you take the absolute value of those numbers, which brings it to a range of zero to two. This is often referred to as the engineering characteristic's absolute marks number.

Then what we do with these marks numbers is we multiply each marked number by performance criteria as weight. As an example, the relationship between maximum speed and time to accomplish task is plus plus. This translates to a two mark number. The weight of time to accomplish task is 13.51%. So, two times 13.51% gives a resulting value of 27.02%. As one more example, the relationship between maximum speed and navigation is X, which remembering that we use the absolute value here translates to a one mark number. That weight and navigation is 10.81. So, one times 10.81 gives resulting value of 10.81. If you add up all the resulting values from these multiplications for each engineering characteristic, you get each engineering characteristics imputed importance. We'll make use of this number more later on.

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But a couple more things to note. For whatever reason, the imputed importance is often presented as a percentage. Now, even though if you do the maths, each engineering characteristic could wind up with an imputed importance of a value of up to 200%, it doesn't really make sense obviously, to represent it as a percentage but just try to think about this as a common and traditional notation.

Now, a more interesting thing to note or an interesting variation if you will, is to actually calculate both the positive and negative imputed importance separately as well. By what we mean by this is we're going to calculate the positive imputed importance using just the positive marks, and we're going to calculate the negative imputed importance by just using the negative marks. The positive and negative imputed importance are still both calculated and presented as a percentage but they may be included in the QFD's basement and within their own rows, or sometimes they are represented in that same spreadsheet's cell with a slash between; positive slash negative. I often find that the positive and negative imputed importance gives you a better measure as to how effective, basically modifying each characteristic is towards bringing benefit to your system.

Now, as a final math check as well, combining the positive and negative imputed importance should add up to your original imputed importance.

Back to Table of Contents

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Activity: Calculate Imputed Importance

In this activity, you will calculate the imputed importance for your system's engineering characteristics.

Task: Calculate the Imputed ImportanceTask: Calculate the Imputed Importance
This is Complete When...This is Complete When...
Worked ExampleWorked Example
Back to Table of Contents

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Watch: Using Competitor Data to Inform Design Targets

You want to account for how your competitors have set their values for the engineering characteristics you have identified. This will help to ensure that design targets you choose will result in a system that is competitive.

In this video, Dr. Schneider describes how to input your competitors' data into the basement of your QFD.

Video Transcript

Now that you have an idea as to the importance of each engineering characteristic, it's interesting to see what our competitors have selected for their values of these engineering characteristics, which can also be very helpful for us in determining our target values for these engineering characteristics. Now, to begin, you want to make sure that you establish what units you want to use when talking about these engineering characteristics. And you want to be consistent in that as well. So when you're talking about meters, you want to use meters all the time; not inches, just use meters. So similarly, when you're talking about things like quality or maybe the algorithm that you're using as those could be different ways we talk about engineering characteristics, we're going to have a common way of talking about those matters as well.

A common way to deal with these, is to create a ranked list of options. So you might have for example a list of all the different kinds of algorithms that you might be considering, where they increase up the list in terms of their complexity and hence perhaps value in producing a good solution to your overall system. Now, once you have the units established, you want to be able to obtain the actual engineering characteristics for your competitors. Now, it's not uncommon actually to buy your competitors and test them. Now, if this is not feasible, due to cost, time or other reasons, an estimate should at least be used because even an estimate is far more informative than nothing.

Now common sources for competitor data can be competitors data sheets or manuals. Sometimes it's third party reviews, sometimes even online forms as you can get enough information can be valuable to collecting this data. But it's also important to distinguish between confident and estimate values. Sometimes with things that you

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are making more an estimate of, you might see them written in italics or different font overall. Now we're trying to think about again as wide of a range of competitors as possible as we talked about earlier. And hence not all engineering characteristics are going to be applicable to all competitors. In these cases, just simply write an N/A for non-applicable, and those cells that don't relate. So if a competitor doesn't have this kind of engineering characteristic, just say N/A.

In this investigation, it's important also to recognize that there could be potentially various constraints as to what could be allowable for various different kinds of engineering characteristics as well. This could be due to government regulations or simply your boss or customers saying that this is the maximum or minimum value you can have for a particular engineering characteristic. Hence these maximum minimum values are also sometimes listed. And in order to distinguish between them, quite often minimum values are often underlined values and maximum values are left in either a normal font or a bold font.

Back to Table of Contents

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Activity: Establish Units, Constraints, and Competitor Values for Engineering Characteristics

In this activity, you will populate a consistent unit of measurement for each engineering characteristic. You will also list the engineering characteristic values for each of the competitors you've identified. Finally, you will acknowledge any additional constraints or requirements for any engineering characteristic.

Task: Input for Comparisons Task: Input for Comparisons
This is Complete When... This is Complete When...
Worked ExampleWorked Example
Back to Table of Contents

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Watch: Assessing Technical Difficulty and Costs

Because technical difficulty and cost can have a significant impact on engineering characteristics, these items also need to be considered when setting your design targets. The QFD basement has rows specifically dedicated to evaluation of these two design factors, and you'll want to fill in this rows before you begin setting design targets.

In this video, Dr. Schneider discusses different factors that might influence technical difficulty and cost, and he shows how to account for both in your QFD.

Video Transcript

When you consider setting targets or making changes to engineering characteristics, it's important to consider both the technical difficulty and cost of making these kinds of decisions. For example, you may want to significantly increase the quality of the path following control system that you have but, this might also be a very technically difficult engineering characteristic to change. Hence, when it comes time to establish your design targets, the technical difficulty measure can help you to consider what you can practically accomplish.

Now, there are many ways to assess technical difficulty and cost. For technical difficulty, some common things that are included can be the technical difficulty to develop the solution for the first time, the difficulty to change the engineering characteristic once the initial work has already been done, the expertise level that is required, perhaps the amount of labor time that would be needed. Sometimes, also just the kind of equipment that would be necessary to use, or any combination of these or other factors. There is no one established standard way to do this. But do remember, this is a design tool, overall, that is meant to help to inform you in making good decisions. You want to choose what is most relevant and helpful for you in determining your technical difficulty.

However, it can be common to see technical difficulty and cost rated on a one through five scale. And it is best whenever possible, to come up with definitions for that one through five scale. Basically meaning, what conditions must be met to earn a one, to earn a two, a three. And some companies do have these standards established already for you. In practice, however, many times this is done far less formally. This is due to the

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fact that often we're working with significant estimates when we're first time we're creating our QFD. And that formality, you know, really may not be as good a fit.

So quite often when you are also establishing the cost estimate for an engineering characteristic, we still use that one through five scale, but again, it's often done as somewhat of an estimate. Now, common costs to consider when assigning a score for an engineering characteristic can include things like fixed cost, variable or non-fixed cost, labor costs, materials costs, equipment costs, including, you know, what it may cost for time to run a special machine. And of course, testing costs because sometimes every time something is changed, it may have to go through an entire validation process again.

As a good practice though, once you have come up with your technical difficulty measure and your cost measure, it's worthwhile to re-evaluate the performance criteria engineering characteristic as well as the engineering characteristics relationships. For example, we might want to try to determine how much it would cost to increase the structural integrity making it out of maybe slightly better or thicker materials. Now, if you look to see that the structural material costs or manufacturing time engineering characteristics are inexpensive enough and technically easy for us to do, we may actually want to reduce the influence structural material cost, the manufacturing time may have on other engineering characteristics. And as an example of that, maximum speed is shown in the QFD roof to have a negative relationship with structural material cost and manufacturing time. But since structural material cost and manufacture time are relatively cheap and easy, we may decide to reduce the negative relationship influence between structural integrity and maximum speed.

Back to Table of Contents

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Activity: Re-evaluate after Technical Difficulty and Cost Considerations

In this activity, you will add estimates for technical difficulty and cost associated with each engineering characteristic. Based on these cost and difficulty estimates, you will then re-evaluate the impact engineering characteristics have on performance criteria and other engineering characteristics.

Task:Re-evaluate after Technical Difficulty and Cost InputsTask: Re-evaluate after Technical Difficulty and Cost Inputs
This is Complete When...This is Complete When...
Worked ExampleWorked Example
Back to Table of Contents

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Watch: Establishing and Iterating Design Targets

One of the main goals for QFD is setting a design target for each engineering characteristic. It may seem that you've been focused on everything but design targets up to this point, but it's with good reason. All the information you've been gathering will help you in the process of creating and refining your set of design targets.

Keep in mind that your ultimate goal is to find targets that give your system the best possible performance. For this reason, you'll begin by adding a new, blank column in the back porch. You'll update the values in this column continually as you populate and adjust each of the design targets. This process is going to be an iterative one.

In this video, Dr. Schneider outlines the process of establishing and fine-tuning your design targets while constantly re-checking the overall performance score. He describes some approaches you might use to set target values, and he lays out a method that works no matter how many engineering characteristics you have. He also discusses the cost/benefit aspect of design target development and explains how managed resources like time and budget can influence your design targets.

Video Transcript

Now it's finally time to attempt our first go at establishing our engineering characteristic design targets. Now even with all that we've done though, this will still take some serious thought to do well. Now, here's a common tip to help monitor your effectiveness at setting your design targets. In your QFD back porch decision matrix, you had basically a column for your current system's performance rating. What I like you do now is add an additional column. Into which you put basically your overall performance score based upon how a system operating at your new design targets would perform. In fact, some people add multiple columns to represent different iterations of different possible combinations of design targets. But the important thing is that we're going to update these new columns as we go.

Now, for setting the actual design targets, there are different approaches to follow, but here's one tried and true way to go. Step one. What I want you to do is to start by filling in the design targets with estimates that you think best represent your current concept that you began your QFD process with. Now, if you don't have these estimates or you no longer like your current concepts estimates because you've gone through

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this QFD process so far, you might want to start with basically your minimum performing engineering characteristic values based upon any constraints or requirements. Or said another way, what must a minimum solution do with regards to engineering characteristics? Now if you don't have any constraints or requirements for those values, you can also use a realistic number inspired by your competitors' engineering characteristic values. And once you have these initial values, make sure again you update the QFD back porch decision matrix with your current estimate of performance so far. Now you can move on to Step two.

Now what we want to do here, is to start with no more than your top five engineering characteristics. We're going to define top as being those with the largest imputed importance. What we want to do is adjust their values using your positive and negative imputed importance to try and improve your estimated overall performance, and update again this change in your new columns in your porch. Now, you may also want to consider both the technical difficulty and costs, and trying to help to figure out what it takes to adjust this engineering characteristic knobs when you're deciding what new values to put. But ultimately, you're going to be asking the question over and over again. How much time and resources you want to dedicate to this engineering characteristic for the potential performance gain you might get from it?

Let me give you an example. If you have an engineering characteristic whose directional change arrow is pointing up, and the positive imputed importance is greater than negative imputed importance, some common guidelines to follow for some initial adjustments could include, increasing the target value until your maximum performance score for adjusting just that one engineering characteristic is achieved. Another way might be to increase the target value to be as large as you can technically accomplish or can afford. And one way to estimate how much you can afford, is perhaps allocating your budget to each engineering characteristic, proportional to your cost score, and its relative imputed importance compared to other engineering characteristics. It's basically just one way of recognizing. You can't spend all your time and budget in one place.

Another way to consider changing engineering characteristics, is to increase the target value to be at least as high as your top competitor. Or if that isn't achievable, perhaps at least, the median value of your competitors. And now again, you go back and make the adjustments to your decision matrix that you have there. Now but once you have a rough idea, on possible design target values for your top engineering characteristics,

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and you've seen the effects on the performance, you can start to address your other engineering characteristics. And again, we recommend that you do this in small batches of roughly five or less. And helping to pick what five you put in that batch, you can continue to use imputed importance, but you may want to consider those of highest technical difficulty or cost at this point as well.

It's also not uncommon that as you go through these additional batches that you might have to readjust some of your past top engineering characteristic target values as you go. You may have recognized that perhaps you spent too much of your budget early on per se. But the key thing is that you continue to make updates to your performance scores as you go, because that is your real measure as to whether, or how well you've really created those design targets.

Back to Table of Contents

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Activity: Assign Design Targets

In this activity, you will evaluate and iterate on your design targets.

Task: Assign Design TargetsTask: Assign Design Targets
This is Complete When...This is Complete When...
Worked ExampleWorked Example
Back to Table of Contents

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Watch: Using QFD to Resolve Counterintutive Results

Ideally, a House of Quality is created at the early stages of design to aid in determining design targets. However, as the concept begins to take shape, it's always good to revisit and iterate through your findings. Even when following your best instinct and the most logical and justifiable approach as you build your QFD, your end result may leave you feeling a little unsure about some of the conclusions you reach based on it.

In this video, Dr. Schneider works through an example to examine a situation in which there is an apparent disconnect between what the House of Quality is saying and what feels right intuitively. He offers some guidance for how to address this kind of disconnect.

Video Transcript

With your QFD completed, it's not uncommon, however, to find surprising results or even results that you may disagree with. So let's return to our rover example and look at kind of an interesting example of setting the maximum speed engineering characteristics design target. Here, the negative imputed importance is actually greater than the positive imputed importance. That says we actually want to decrease the speed to improve our overall performance. Now this may seem very counterintuitive because if we want to accomplish more tasks, we need to go faster, right?

Well, all right, so what do you do in these cases? First thing you do, is you remember that the QFD body, which is where we calculate the imputed importance, shows again, the relationship of the engineering characteristics to performance criteria. And if we again look here, we can find, yes, I do believe we have filled this out well, and it does actually match what is most important for our system.

All right. Let's examine another part of the QFD, maybe the roof. Here we investigate again engineering characteristics' influence on other engineering characteristics. And we may recognize that all right, changing the speed could actually make it harder to make various different changes to other important engineering characteristics. Okay, well that starts to make sense and it looks like we filled that part out correctly but still, lowering the maximum speed sounds really counterintuitive.

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All right, well let's at least now just start to follow along what the QFD has generated and see what results could come out with regards to performance. Okay. So we want to lower the speed. As your max speed vehicle, however, may not be the wisest or practical solution. So instead, perhaps we'll begin with setting the design target at the lowest max speed that we're comfortable with, which is perhaps the lowest our competitors have, or the lowest that our constraints or requirements will allow. Okay and then this should be able to improve the overall performance. And if we look at how we're able to adjust all the engineering characteristics with a lower speed, we see that does actually improve the performance.

But wait a second, we still have this natural feeling that we really want the vehicle to go fast. Why is that? We're having a hard time rationalizing this. Well, part of the reason may be that the maximum speed has a very positive relationship with the most important and most heavily weighted performance criteria, which is trying to accomplish tasks. It has come that we as humans, we naturally tend to optimize around where we see the greatest gain, often considering only a few factors. In doing so, we may naturally ignore the importance of other factors and it may in cases result in an overall sub-optimal, or even potentially invalid solution. Even though we feel it's great, and we feel like we'll be able to defend why. Well of course this is a great solution. Look how good it does at our top performance criteria.

Well, what makes this situation even more challenging though, is that the maximum speed has a very strong influence on the coolness performance criteria. We as humans are naturally more susceptible and give more weight to things like coolness or more generally anything that has the kind of perceived value. Otherwise marketing wouldn't work. But we should have already thought about how we're going to measure and how much weight to give to any perceived value in a performance metrics. So we should have already taken this into account. And if the coolness solution still isn't winning, we may need to fight our own intuition.

Instead, we have to constantly remind ourselves of the needs of our challenge and how the performance of any solution will be evaluated as a whole. Now, when you make you can look at every design as kind of like its own kind of game. And how to best position all the engineering characteristics in your effort in order to gain the highest score. Just make sure that what you're scoring and you're scoring your design, is by the challenges performance criteria, not just your own. The QFD offers a formal process, to help us question ourselves, allow us to redefine our understanding of any challenge.

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In this video, Dr. Schneider shares how the team who worked on the example in the first video benefited from their House of Quality insight.

Video Transcript

You want to know one of the best parts of this robotic vehicle example is? This is not just an example that has been contrived to help make an important point, its actually taken from part of a larger QFD that was done for real world project. And yes, aiming for the system to run just above the minimum required speed, proved to be a very wise decision. This decision was able to come to by recognizing the negative impact of faster speeds on performance criteria such as navigation and safety in particular. In turn, this helped the team to recognize, how they were basically making the problem far harder on themselves by trying to go fast. As essentially, developing many of the engineering characteristics for the real robot vehicle, related to autonomous navigation, particularly the path falling control system, turned out to be very complicated. Had a faster system been designed, it would have made these engineering characteristics even harder to develop. And it is unlikely that a working solution would have been achieved in the time allowed. Instead, the team was able to develop a system that was easier to test and learn from, and still met all the challenges needs very well. Enough so, that the team even later earned a substantial contract, to create the next version of the robot. The team was then able to reiterate on the design but now, they had a better understanding of the supporting subsystems and challenges needs, that would arise from going faster. And hence, they were able to lessen the negative impact that going faster would have on key performance criteria like navigation, safety and structural integrity as well. This shifted basically the positive imputed importance of going faster, to be significantly greater than going slower and hence, going faster, was a more well justified decision, in the later iteration.

Back to Table of Contents

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Discussion: Design Target Justifications

You have come a long way in building out your QFD. At this point, you have developed the best estimates of performance that you currently can based on the design targets you have selected. In some cases, you may have been at a loss to determine what a specific design target should be. And in some cases, you may still feel that a result is counterintuitive. In any case, though, QFD has allowed you to consider multiple factors while coming up with a justifiable best initial target for each characteristic.

Instructions:

You are required to participate meaningfully in the course discussions.

Discussion topic:

Consider your system's engineering characteristics and choose one you feel is important for your design and for which you feel you have a strong justification behind your initial design target. Create a post in which you:

- briefly describe your system concept and the engineering characteristic you have chosen to focus on for this discussion.
- indicate why you think this engineering characteristic is important for your system (hint: you may wish to discuss why it was given a high imputed importance score.)
- explain your justification for the design target and describe any reasoning as well as any revision you went through to arrive at the target.
- if your target creates a counterintuitive result, explain why the result is counterintuitive and why you feel you should stick with it anyway, at least for now.

Once you have created your post, choose at least one post made by another student and create a reply in which you:

- ask clarifying questions about the design target
- share any lessons learned from reading the original post

To participate in this discussion:

Click **Reply** to post a comment or reply to another comment. Please consider that this is a professional forum; courtesy and professional language and tone are expected.

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Before posting, please review eCornell's policy regarding plagiarism (the presentation of someone else's work as your own without source credit).

Please note:

While discussion board postings will be accepted through the end of the course, we strongly encourage everyone to move through this course as a group. As such, postings that are made to this board after 5pm ET on the due date will be read and graded but may not receive a response on the board from your facilitator and/or your peers. Please let your facilitator know if you have any questions.

Back to Table of Contents

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Assignment: Course Project, Part Two—Quality Functional Deployment

In this part of the project, you will build out the basement of your House of Quality.

In the basement you will collect additional information about your engineering characteristics, including:

- the set measurement units
- the engineering values of your competitors
- any external requirement thresholds
- the imputed importance
- any technical difficulty
- any estimated cost

With all this information established, you then will provide educated estimates toward the **design targets** for your system.

In addition to filling out the basement, you will create a new column on the back porch and fill it out with a set of estimated scores based on a system that is designed with the targets you have just added in the QFD basement.

Completion of all parts of this project is a course requirement.

Instructions:

1. Open your saved course project document. (If needed, **download** it again now.)
2. Complete Part Two.
3. Save your work as one of these file types: .doc, .docx, .txt, .pdf, .xls, .xsls. No other file types will be accepted for submission.
4. Now you will submit your completed course project and your completed QFD workbook for review and credit.
5. Click the **Submit Assignment** button on this page to attach your completed course project document and QFD workbook and send them to your instructor for evaluation and credit.

Do not hesitate to contact your instructor if you have any questions about the project. You will add to this document as the course proceeds and will submit it to the course

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instructor at the end of the course.

Before you begin:

Before starting your work, please review the **rubric** (a list of evaluative criteria) for this assignment. Also review eCornell's policy regarding **plagiarism** (the presentation of someone else's work as your own without source credit).

Back to Table of Contents

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Module Wrap-up: Define and Refine Design Targets

In this module, you determined initial targets for engineering characteristics. You also calculated performance estimates for a system based on your initial design targets. In this way, you have forged a connection between the engineering characteristics for your initial design and the performance measures your customers care about.

Now with a QFD that expresses relationships between engineering characteristics and other engineering characteristics as well as these characteristics' impact on performance, you can communicate the relationships and examine trade-offs at the level of the individual characteristic. And you can argue effectively that your chosen design targets will lead your team to a winning solution.

Back to Table of Contents

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Thank You and Farewell

Congratulations on completing *Implementing the Quality Function Deployment Method*. As you have seen, determining initial targets for your design parameters is important in achieving a great overall design.

From all of us at Cornell University and eCornell, thank you for participating in this course.

Sincerely,

Professor David R. Schneider

David R. Schneider
Senior Lecturer
College of Engineering
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